

Étale Coverings and the Fundamental Group

(SGA 1)

English source-synchronization workpass

Exposé I opening through § I.2

Task-owned working translation

18 July 2026

Source and status note

This bounded workpass translates the opening of Exposé I through §I.2 directly from the French T_EX source in arXiv `math/0206203v2`, with archive SHA-256:

```
0A33C9C06908705A2525690FAAE02F6F
07980A4AA069A8B6CFF9B1D9BC39ACD3
```

The current translated source boundary is `smf_doc-math_3_01.tex`, lines 556–652; line 653 begins §I.3 and is outside the current workpass.

Recovered English materials associated with jcreinhold, jmoellermath, and Tim Hosgood were consulted only as target-language comparison controls. They are not source authority, and no agreement among them was treated as proof of source alignment. The two Hosgood-related repositories belong to one derivative lineage, not two independent witnesses.

This is a source-audited working translation of one bounded unit. It is not a complete SGA 1 translation, a critical edition, or a certification of the mathematics. Rights and attribution for any public redistribution remain subject to archive-maintainer review; the French source header asks modified versions to identify themselves, which this workpass does explicitly.

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CHAPTER I

Étale morphisms

To simplify the exposition, we assume that all preschemes under consideration are locally noetherian, at least after no. I.2.

French source
p. 1

I.1. Notions of differential calculus

Let X be a prescheme over Y , and let $\Delta_{X/Y}$, or simply Δ , be the diagonal morphism $X \rightarrow X \times_Y X$. It is an immersion, hence a closed immersion of X into an open subset V of $X \times_Y X$. Let $\mathcal{I}_X$ be the ideal of the closed subscheme corresponding to the diagonal in V (N.B. if one wishes to proceed intrinsically, without assuming that X is separated over Y —a hypothesis that would be farcical—one should consider the set-theoretic inverse image of $\mathcal{O}_{X \times X}$ in X , and denote by $\mathcal{I}_X$ the augmentation ideal in the latter...). The sheaf $\mathcal{I}_X/\mathcal{I}_X^2$ may be regarded as a quasi-coherent sheaf on X ; we denote it by $\Omega_{X/Y}^1$. It is of finite type if $X \rightarrow Y$ is of finite type. It behaves well under base change $Y' \rightarrow Y$. We also introduce the sheaves $\mathcal{O}_{X \times_Y X}/\mathcal{I}_X^{n+1} = \mathcal{P}_{X/Y}^n$; these are sheaves of *rings* on X , making X into a prescheme that may be denoted by $\Delta_{X/Y}^n$ and called the *n-th infinitesimal neighborhood of X/Y* . The resulting sorites is entirely trivial, although rather long¹; it would be prudent to speak of it only when there is something useful to say about it, in connection with smooth morphisms.

I.2. Quasi-finite morphisms

PROPOSITION I.2.1. *Let $A \rightarrow B$ be a local homomorphism (N.B. the rings are now noetherian), and let $\mathfrak{m}$ be the maximal ideal of A . The following conditions are equivalent:*

- (i) $B/\mathfrak{m}B$ is finite-dimensional over $k = A/\mathfrak{m}$.
- (ii) $\mathfrak{m}B$ is an ideal of definition, and $B/\mathfrak{r}(B) = \kappa(B)$ is an extension of $k = \kappa(A)$.
- (iii) The completion $\hat{B}$ is finite over the completion $\hat{A}$ of A .

One then says that B is *quasi-finite* over A . A morphism $f: X \rightarrow Y$ is said to be quasi-finite at x (or the Y -prescheme f is said to be quasi-finite at x) if $\mathcal{O}_x$ is quasi-finite over $\mathcal{O}_{f(x)}$. This is also equivalent to saying that x is *isolated in its fiber* $f^{-1}(f(x))$.² A morphism is said to be quasi-finite if it is so at every point.³

French source
p. 2

COROLLARY I.2.2. *If A is complete, quasi-finite is equivalent to finite.*

One could give the usual chain of implications (i), (ii), (iii), (iv), (v) for quasi-finite morphisms, but this does not seem indispensable here.

¹Cf. EGA IV 16.3.

²The original printing (Exposé I, p. 2) and both French arXiv renderings print $f^{-1}(x)$. Since $x \in X$ for $f: X \rightarrow Y$, the type-correct fiber through x is $f^{-1}(f(x))$, which is used here with the source defect explicitly disclosed.

³In EGA II 6.2.3 one assumes in addition that f is of finite type.